

Reactor capital cost governs small modular reactor cogeneration for data-center power and cooling

Honglin Li^{1,3}, Buxin She³, Jie Zhang^{1,2*}

¹Department of Mechanical Engineering, The University of Texas at Dallas, Richardson, 75080, TX, USA.

²Department of Electrical and Computer Engineering, The University of Texas at Dallas, Richardson, 75080, TX, USA.

³Department of Electrical and Computer Engineering, Kansas State University, Manhattan, 66506, KS, USA.

*Corresponding author(s). E-mail(s): jiezhang@utdallas.edu;

Abstract

Data centers are becoming large, continuous electricity loads, but the value of colocating them with small modular reactors that also recover heat for cooling remains uncertain under current market and policy conditions. Here we co-optimize electricity supply, steam extraction, absorption cooling, grid exchange and storage for a 200 MW_e data center in the Texas power market over 100 hourly scenarios spanning 2022–2024 prices, reactor capital costs, financing, carbon prices, cooling efficiency and scale. At 2023 mid-range reactor capital, the nuclear configurations cost 146–149% more than grid supply even with the nuclear production tax credit, whereas nth-of-a-kind reactor capital makes them 22–25% cheaper without a carbon price. Absorption cooling supplies 94% of annual cooling but changes cost by only \$2.4 M yr⁻¹, showing that reactor capital, not heat recovery, sets the viability boundary.

Keywords: Small modular reactor, Absorption cooling, Data center, Techno-economic analysis, Mixed-integer linear programming, Thermal integration

1 Introduction

Data centers are emerging as a major driver of electricity-demand growth after a decade in which efficiency gains moderated their global energy use¹. U.S. data-center

electricity consumption nearly doubled between 2019 and 2023 and is projected to reach 6.7–12% of total U.S. electricity by 2028²; globally, electricity use from data centers is projected to roughly double to 945 TWh by 2030^{3,4}. In the Electric Reliability Council of Texas (ERCOT), large-load interconnection requests already exceed the scale of conventional planning margins^{5,6}. This growth creates a supply problem that is not only about annual energy: hyperscale campuses are continuous loads, require high availability, and carry cooling demands that rise sharply during hot and humid hours^{7,8}.

Small modular reactors (SMRs) have therefore re-entered data-center procurement discussions as firm, low-carbon generators whose unit sizes are closer to a hyperscale campus than conventional gigawatt-scale nuclear plants^{9–12}. Yet the economic boundary remains unsettled. Prior techno-economic assessments show that SMRs can be lower-risk to build than large reactors but often remain above grid supply without a carbon constraint¹². Capital-cost estimates also span a wide first-of-a-kind (FOAK) to nth-of-a-kind (NOAK) range, with learning, standardization, financing and project delivery determining whether reactors approach published low-cost trajectories^{11,13,14}. Current U.S. policy adds a second uncertainty: the Section 45U nuclear production tax credit (PTC) can materially change the cost of operating a qualifying nuclear plant, but only if the pre-credit gap is small enough for the credit to close.

Cooling-side thermal integration offers a different route to value. Data-center cooling can account for 20–40% of facility energy use, and absorption chillers can convert heat into chilled water when the heat source is sufficiently hot and available^{15–17}. Reactor steam cycles can provide the driving temperatures required by double-effect lithium-bromide absorption chillers more readily than server exhaust, but the value of this integration depends on chiller capital cost, extracted-steam opportunity cost, wet-bulb-driven de-rating and backup electric cooling during crystallization-risk hours^{18–20}. Treating heat recovery as an automatic benefit therefore risks overstating the value of nuclear cogeneration for an efficiently cooled campus.

The remaining gap is a boundary problem. Existing studies rarely combine hourly cooling physics, multi-year electricity and gas prices, reactor capital-cost trajectories, production-tax-credit policy and carbon accounting in one co-optimization. One-dimensional sensitivities also make it difficult to tell whether a proposed site or contracting structure lies in a competitive region. The needed result is not a single leveled cost but a multi-axis map: when does a colocated reactor beat grid procurement, when does absorption cooling actually add value, and which assumptions move the boundary?

Here we present a techno-economic boundary analysis of a 200 MW_e data center colocated with a BWRX-300 SMR and a double-effect absorption chiller driven by cascaded high-pressure to low-pressure turbine extraction. A mixed-integer linear program co-optimizes power supply, cooling, grid exchange and storage at hourly resolution across 2022–2024 ERCOT prices, Henry Hub gas prices and Houston wet-bulb temperatures, with the Section 45U PTC applied to the nuclear cases. We compare grid supply, reactor supply without heat recovery, reactor supply with absorption cooling, and an on-site natural gas combined cycle (NGCC) across 100 scenarios spanning data-center scale, cooling efficiency, market year, battery deployment, reactor and

absorption capital cost, carbon price and weighted-average cost of capital (WACC); the case definitions, optimization formulation and scenario ledger are specified in Methods and the Supplementary Information. We find that nuclear cogeneration is a conditional strategy: at mid-range reactor capital it remains far above grid cost, but at NOAK reactor capital the PTC opens a viable region with no carbon price, while absorption cooling supplies most cooling yet does not set the economic boundary.

2 Results

2.1 A boundary analysis separates supply, heat recovery and policy effects

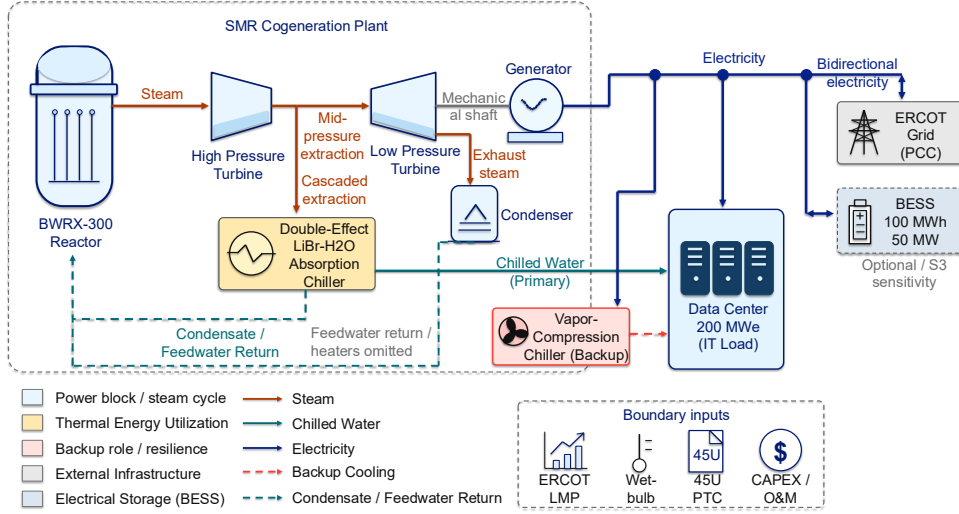


Fig. 1 Cascaded steam-extraction mechanism for SMR-powered data-center cooling. Steam from the BWRX-300 power block expands through the high-pressure turbine, a mid-pressure extraction stream drives a double-effect LiBr-H₂O absorption chiller, and the remaining steam expands through the low-pressure turbine to produce electricity. Chilled water supplies the data center, vapor-compression cooling provides backup during absorption limits, and the grid and battery interfaces balance electricity. Complete case definitions, optimization equations and scenario groups are provided in Methods and the Supplementary Information. SMR, small modular reactor; LiBr, lithium bromide; ERCOT, Electric Reliability Council of Texas; PCC, point of common coupling; BESS, battery energy storage system; CAPEX, overnight capital expenditure; O&M, operations and maintenance.

The analysis is organized around a single competitiveness metric, the grid-cost margin, defined as one minus the ratio of each configuration's total annualized cost (TAC) to the matching grid-only data-center cost. Positive values mean that an on-site option beats grid procurement; negative values mean that it does not. This framing matters

because the reactor, cooling system and policy support can move the margin in different directions. Reactor capital and financing determine the annualized fixed-cost burden, absorption cooling changes both chiller electricity and turbine output, grid exchange determines merchant-export value, and the Section 45U PTC offsets eligible nuclear generation.

Figure 1 presents the fully integrated Case 2 configuration as the organizing system schematic, with reactor power conversion, cascaded steam extraction, absorption cooling, vapor-compression backup, grid exchange and optional storage shown in one architecture. The four model configurations are then defined as supply and cooling variants of this architecture in Methods and Table 1, the hourly optimization workflow is specified by the formulation in Methods, and the scenario envelope is given in the scenario-design subsection and Supplementary Table S7. The Results first establish the baseline economic gap, then ask whether operations, cooling-side value, reactor capital, financing, market year or carbon policy can close it.

2.2 Mid-range reactor capital keeps nuclear options above grid supply

At the 2023 National Laboratory of the Rockies (NLR) Annual Technology Baseline mid-range (ATB-Mid) reactor-capital baseline, the nuclear configurations are not close to grid parity. The grid-supplied data center (Case 0) has a TAC of \$78.1 M yr⁻¹, of which 93% is the electricity bill. The two reactor configurations cost \$192.2 and \$194.6 M yr⁻¹ without heat recovery (Case 1) and with absorption cooling (Case 2), respectively, even after the Section 45U PTC returns \$26.4 and \$25.8 M yr⁻¹ (Figure 2a). Reactor capital recovery dominates both totals. Grid-export revenue and the PTC offset part of the fixed-cost burden, but not enough to prevent grid-cost margins of -146% and -149%.

The only baseline configuration that beats grid procurement is the on-site NGCC plant (Case 3), at \$55.7 M yr⁻¹ and a +29% grid-cost margin. This result is tied to the 2023 fuel-price regime: delivered natural gas averaged \$2.63 MMBtu⁻¹, while ERCOT day-ahead prices remained high enough for the grid-only bill to exceed on-site fuel cost. Because Case 3 is islanded, it cannot import during the roughly 42% of hours when the day-ahead price falls below its running cost, so the comparison is an annual cost result rather than an hour-by-hour arbitrage result. Adding absorption cooling to the nuclear plant changes cost by only \$2.4 M yr⁻¹ relative to the reactor-only case, showing already at the baseline that heat recovery is not the primary economic lever.

The BWRX-300 unit-size constraint explains part, but not most, of the baseline nuclear penalty. The smallest commercial unit is 270 MW_e net at the assumed capacity factor, about 2.9 times the data center’s 94.7 MW_e average information-technology (IT) load, so Case 2 exports about 1.43 TWh yr⁻¹ to ERCOT. Scaling the campus to 2.5× the baseline load brings the nuclear site close to import-export balance, but the matched Case 2 margin is still -62%; at 3.0× the site becomes a net importer and the margin improves only to -52% (Supplementary Figure S3). Unit-size matching therefore improves utilization intensity, reducing the Case 2 excess cost from \$140 to \$49 MWh_{IT}⁻¹ across the sweep, but it does not remove the ATB-Mid reactor-capital gap.

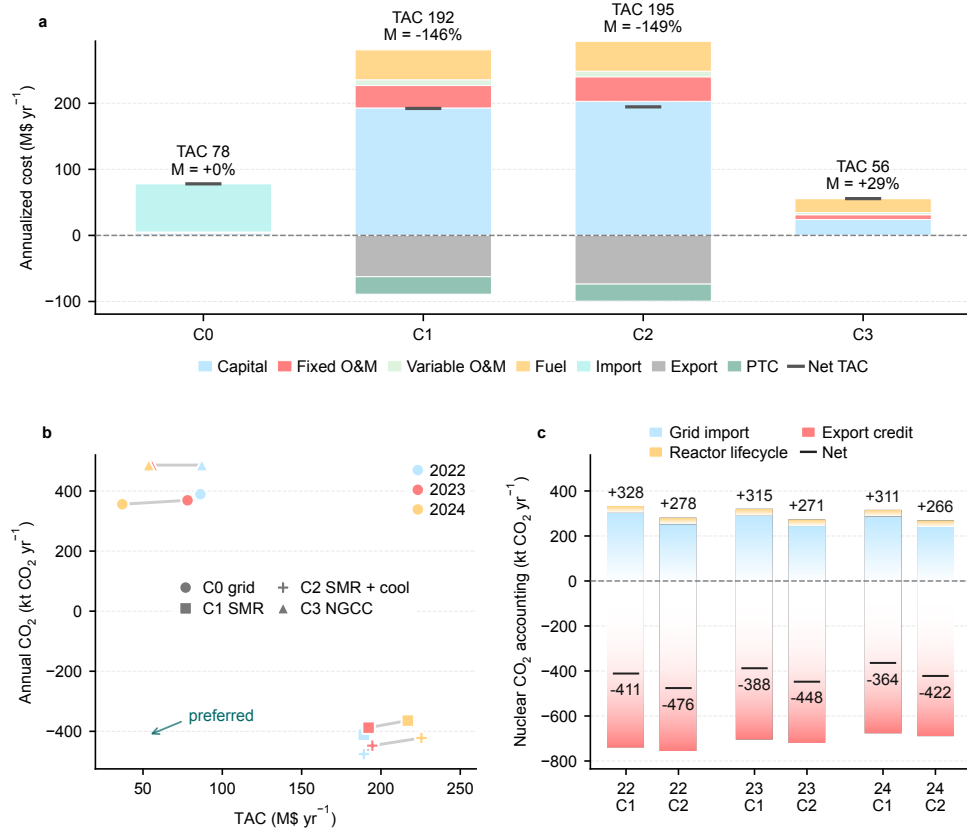


Fig. 2 Baseline economics of the four configurations at the 2023 ATB-Mid baseline. ATB-Mid denotes the mid-range National Laboratory of the Rockies Annual Technology Baseline reactor-capital case; SMR, small modular reactor; NGCC, natural gas combined cycle; PUE, power usage effectiveness; WACC, weighted-average cost of capital; O&M, operations and maintenance; PTC, production tax credit; TAC, total annualized cost; ERCOT, Electric Reliability Council of Texas. Baseline assumptions are full-load PUE 1.35 and WACC 6.7%. (a) TAC decomposition: each bar stacks capital recovery, fixed and variable O&M, fuel and grid imports; bars below zero are grid-export revenues and, for the nuclear cases, the Section 45U PTC; the horizontal mark is the net TAC and M is the grid-cost margin against Case 0. (b) Cost-carbon trade-off across ERCOT years 2022, 2023 and 2024: marker shape denotes case, marker color denotes year, and the arrow indicates the Pareto-preferred direction. (c) Carbon-accounting components for the two nuclear configurations: grid-import and reactor-lifecycle emissions are offset by grid-export credits, and the horizontal mark reports net annual CO₂.

The baseline cost ranking is mirrored in the cost-carbon plane (Figure 2b,c). The nuclear cases occupy the low-carbon region after reactor-lifecycle emissions and grid-export credits are accounted for using the ERCOT hourly average carbon intensity, while the grid-only case anchors the low-cost region in 2023 and 2024. Case 3 trades higher emissions for lower cost under 2023 gas prices. Figure 2c shows that export credits more than offset grid-import and reactor-lifecycle emissions in Cases 1 and

2, but low carbon intensity alone does not make either reactor configuration cost-competitive at zero carbon price and mid-range reactor capital.

2.3 Absorption cooling supplies most cooling but adds little value

The hourly dispatch of Case 2 shows how the absorption chiller integrates with the reactor under winter and summer operating conditions (Figure 3a,b). In the winter week, the BWRX-300 runs near nameplate thermal output, consistent with its 92% annual capacity-factor constraint. The high-pressure to low-pressure turbine train produces close to its full $\sim 270 \text{ MW}_e$ net output through most of the window, turning down to about 125 MW_e during the low-price hours. Chilled-water demand tracks the 25–142 MW_e winter IT load through $Q_{\text{cool}}^{\text{dem}} = P_{\text{IT}}/\eta_{\text{ch}}$, and the extraction tap supplies most of it as steam, with the VCC backup contributing only on the highest-demand hours. In the July week, wet-bulb temperature climbs toward the 26°C design anchor on the warm afternoons, derating the absorption coefficient of performance (COP) toward its 1.10 design value. The week stays below the 27°C crystallization gate (32°C cooling-water inlet). The higher summer cooling load nonetheless exceeds what the derated absorber can supply from steam, so the VCC backup tops up to about 30 MW_c on the warmest afternoons. Across the full year the absorption chiller delivers 862 GWh_c and the VCC backup 60 GWh_c , the latter spread over roughly 2,870 hours of partial top-up that also cover the 5% maintenance derating of the absorber (Supplementary Information). The crystallization gate itself is rare: the availability indicator in Equation 8 forces the absorber fully offline for about 115 hours, concentrated in late June through August (annual peak wet-bulb 28°C), during which the backup carries the entire cooling load of those hours (13.6 GWh_c).

The waterfall in Figure 3c decomposes the $\$2.4 \text{ M yr}^{-1}$ gap between Case 2 and Case 1 into the marginal components of adding absorption cooling to the nuclear-only plant. The chiller displaces $\$18.2 \text{ M yr}^{-1}$ of gross vapor-compression electricity at the ERCOT 2023 hourly import price. Against this credit stand, in annual terms, $\$13.2 \text{ M}$ for absorption capital plus fixed operations and maintenance, $\$4.1 \text{ M}$ for the Willans-line electricity opportunity cost of extracted steam, $\$2.3 \text{ M}$ for VCC backup and top-up, and $\$0.6 \text{ M}$ for the reduction in the production tax credit, since diverting steam to the absorber lowers the creditable turbine output; the monetized direct-site-water increment is only $\$0.04 \text{ M yr}^{-1}$. A small $-\$0.3 \text{ M yr}^{-1}$ residual reconciles the component sum with the measured TAC difference. Credit and cost nearly cancel, so the chiller is close to break-even, and the residual penalty is a capital-cost problem rather than one of operating physics or water.

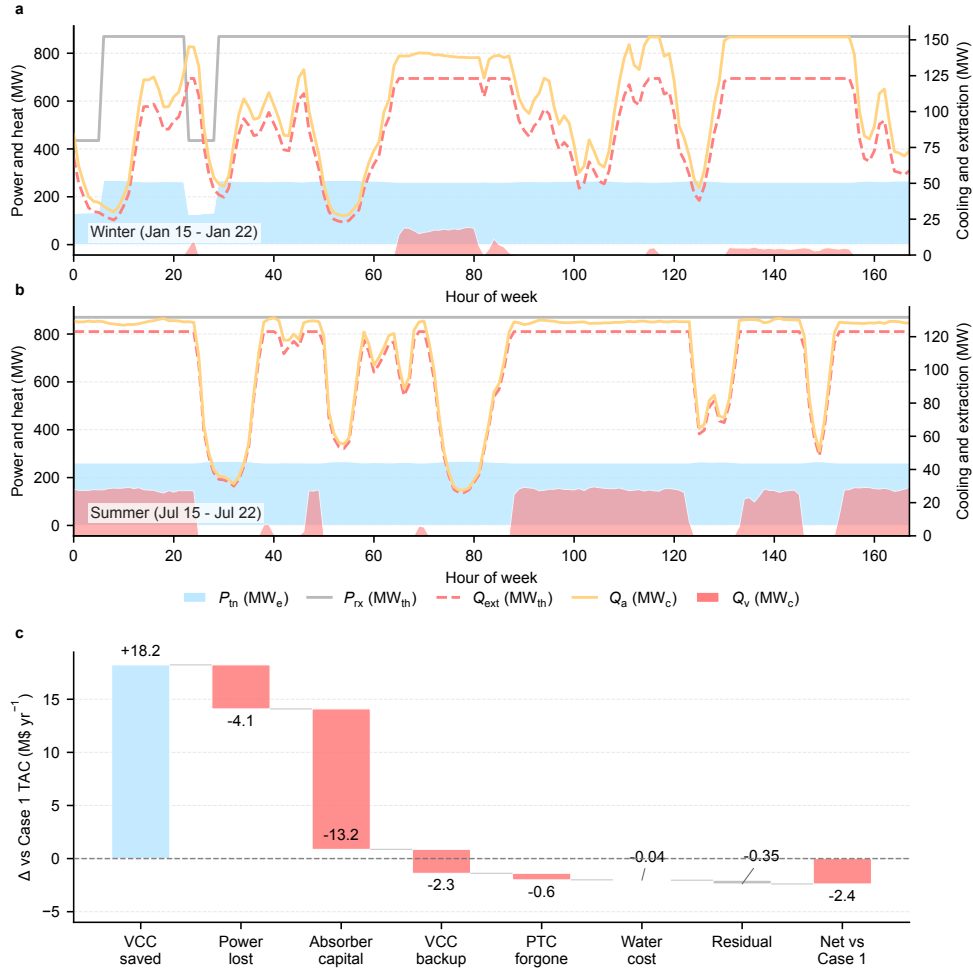


Fig. 3 Cogeneration operation and the marginal value of absorption cooling for Case 2. VCC denotes vapor-compression chiller; TAC, total annualized cost; PTC, production tax credit. (a) Hourly dispatch over a winter week (15–22 January) and (b) a summer week (15–22 July): reactor thermal output P_{rx} and turbine net electric output P_{tn} are read on the left axis; mid-pressure extraction Q_{ext} , absorption cooling Q_a , and VCC backup cooling Q_v on the right axis. In both plotted weeks the wet-bulb stays below the 27 °C crystallization gate, so the absorption chiller carries most of the cooling load; the VCC backup tops up on the warmest summer afternoons, when the derated absorber cannot meet demand from steam, and takes over entirely during the roughly 115 crystallization-gate hours yr⁻¹ elsewhere in the record. (c) Waterfall decomposition of the marginal annual value of adding the absorption chiller to a nuclear-only plant (Case 2 against Case 1): positive bars reduce Case 2 TAC, negative bars increase it, VCC backup covers steam-limited top-up and crystallization-gate fallback, PTC forgone is the production tax credit lost as steam diverts from the turbine, and the residual bar reconciles the component sum with the measured \$2.4 M yr⁻¹ difference.

148 With the heat load fixed at P_{IT}/η_{ch} , the cooling-side lever is the efficiency of the
149 electric chiller rather than the size of the duty. The cooling-efficiency sweep (Supple-
150 mentary Figure S2) varies the chiller coefficient of performance so that the reported
151 power usage effectiveness (PUE) spans $\{1.10, 1.30, 1.50\}$ for Cases 1 and 2, with all
152 other parameters at the 2023 ATB-Mid baseline. The absorption chiller delivers a near-
153 constant $862 \text{ GWh}_c \text{ yr}^{-1}$ across the sweep; what changes is the value of the electric
154 cooling it displaces. Net cooling-electricity saving after VCC backup and top-up rises
155 from $\$1.9$ to $\$13.8$ and $\$23.0 \text{ M yr}^{-1}$ as PUE increases from 1.10 to 1.30 and 1.50. As
156 a result, Case 2 costs $\$13.0 \text{ M yr}^{-1}$ more than Case 1 when the electric chiller is highly
157 efficient, narrows to a $\$4.6 \text{ M yr}^{-1}$ penalty at PUE 1.30, and becomes a $\$4.6 \text{ M yr}^{-1}$
158 saving at PUE 1.50. The main-results baseline uses a COP-3.2 electric chiller at a full-
159 load PUE of 1.35, so its $\$2.4 \text{ M yr}^{-1}$ Case 2-minus-Case 1 gap sits just below the
160 break-even point near PUE 1.40.

161 2.4 Viability opens only along the low reactor-capital frontier

162 The ERCOT price regime during the year of operation is the largest single mar-
163 ket sensitivity (Figure 4a). The 2022 cluster captures the high natural-gas price
164 regime: delivered natural gas (Henry Hub plus Houston Ship Channel basis) averaged
165 $\$6.53 \text{ MMBtu}^{-1}$, and the NGCC margin falls to -1% , essentially parity with the grid,
166 because the grid clearing price and the on-site fuel cost moved together. In 2023, low
167 gas prices ($\$2.63 \text{ MMBtu}^{-1}$) combined with high ERCOT prices made Case 3 the
168 only on-site option that beat grid supply (a $+29\%$ margin). The 2024 cluster shows
169 the deepest negative margin for every on-site option because ERCOT LMPs settled
170 at a multi-year low, an annual mean of about $\$28 \text{ MWh}_e^{-1}$, making grid procurement
171 comparatively inexpensive. Across the three years, no on-site option beats grid supply
172 in 2024 or 2022, and only NGCC beats it in 2023. The year regime therefore sets the
173 market boundary for the on-site gas option: a $\pm 50\%$ swing in Henry Hub gas price or
174 ERCOT LMP can overturn the ranking between grid and NGCC supply. Nuclear, by
175 contrast, stays far above grid cost in all three years, so its ranking is governed by the
176 capital-cost rather than the market axis.

177 The battery energy storage system (BESS) axis tests whether a 100 MWh,
178 50 MW lithium-ion battery changes the competitiveness frontier. It lowers TAC by
179 $\$0.63 \text{ M yr}^{-1}$ in each nuclear case, improving the margin by only 0.8 percentage
180 point and never altering the configuration ranking (see Supplementary Information).
181 Because this effect is much smaller than the market, capital-cost and financing axes,
182 it is retained as a secondary boundary rather than as a main result.

183 SMR overnight capital cost is the dominant technology-cost axis (Figure 4b). At
184 FOAK ($\$14,700 \text{ kW}_e^{-1}$) the margin is -372% (Case 1) and -375% (Case 2): reactor
185 capital outweighs all other costs by roughly a factor of five, and the production tax
186 credit is far too small to matter. At ATB-Mid the margin settles at -146% (Case 1)
187 and -149% (Case 2), so even with the credit, mid-range capital leaves the nuclear
188 cases about 2.5 times more expensive than grid supply. At NOAK ($\$2,250 \text{ kW}_e^{-1}$)
189 the picture inverts: the margin rises to $+25\%$ for Case 1 and $+22\%$ for Case 2, so
190 both nuclear configurations are cheaper than grid procurement with no carbon price
191 applied. The credit is decisive precisely here, because the pre-credit gap to grid at

192 NOAK capital is small, on the order of 10%, and the \$26.4 M yr⁻¹ Case 1 credit more
 193 than covers it, whereas the same credit barely dents the far larger FOAK and ATB-
 194 Mid gaps. Coupling the NOAK reactor with the absorption chiller (Case 2 NOAK) is
 195 marginally less competitive than Case 1 NOAK, because the absorption capital still
 196 slightly outweighs the cooling-side savings, but both stay comfortably below grid cost.
 197 The financing assumption changes the same capital-dominated boundary rather
 198 than creating a separate operating mechanism (Figure 4c). Sweeping the weighted-
 199 average cost of capital between 5% and 10% for Case 2 while holding absorption
 200 capital, reactor capital and operating parameters fixed moves TAC from \$168 to
 201 \$251 M yr⁻¹, an \$83 M yr⁻¹ range. The implied grid-cost margin ranges from -115%
 202 at 5% WACC to -221% at 10% WACC. By comparison, the full range of group G4
 203 (reactor capital, FOAK to NOAK) moves the Case 2 margin between -375% and
 204 +22%, crossing the parity line. Financing materially affects the high-cost FOAK-ATB
 205 region but matters less than the reactor capital-cost axis once the trajectory reaches
 206 NOAK. Policy or contracting mechanisms that lower the effective cost of capital for
 207 SMR projects, such as long-term offtake agreements, sovereign loan guarantees, or
 208 accelerated depreciation, can compress but not eliminate the gap to grid supply.

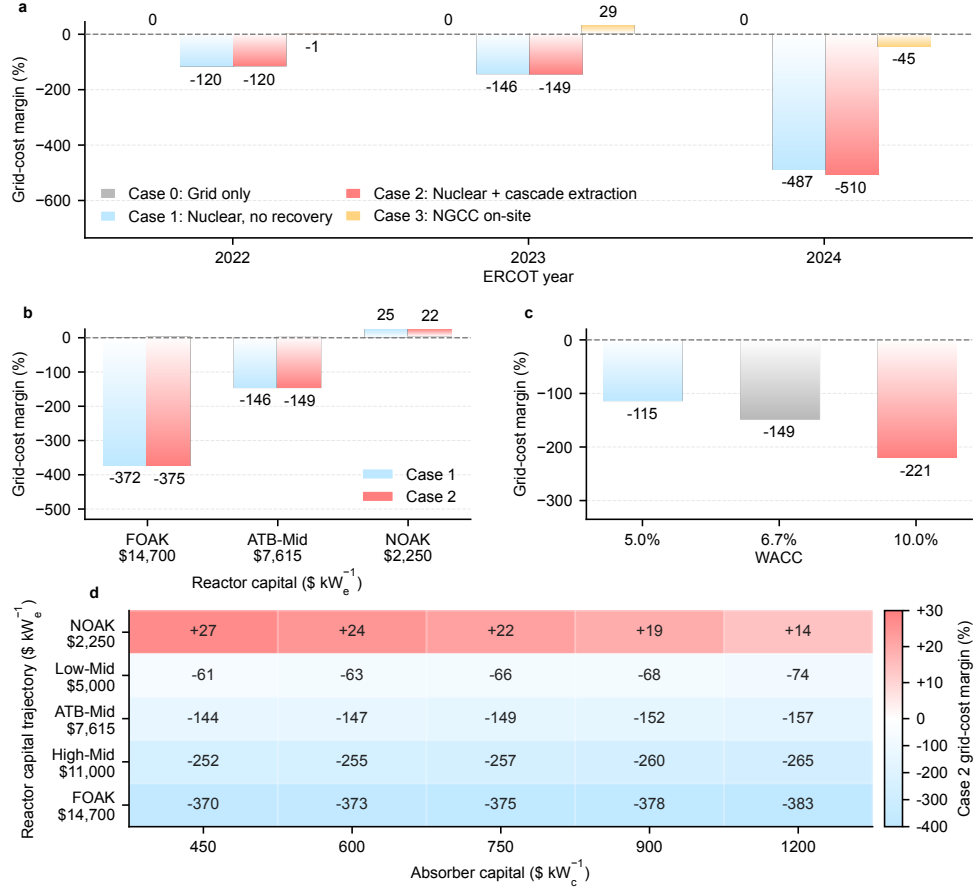


Fig. 4 Market, reactor-capital, financing and joint capital-cost boundary atlas. ERCOT denotes Electric Reliability Council of Texas; NLR ATB, National Laboratory of the Rockies Annual Technology Baseline; WACC, weighted-average cost of capital; SMR, small modular reactor; FOAK, first-of-a-kind; NOAK, nth-of-a-kind. (a) Grid-cost margin for the four configurations across ERCOT years 2022, 2023 and 2024. (b) Grid-cost margin for Cases 1 and 2 across the NLR ATB 2024 BWRX-300 capital-cost trajectories. (c) Case 2 grid-cost margin across the WACC sweep. (d) Joint SMR and absorption-chiller capital-cost viability frontier for Case 2; cell values are grid-cost margin against the 2023 grid-only baseline. Bars or cells below zero indicate configurations that are more expensive than the grid-only baseline.

The joint capital-cost frontier in Figure 4d turns the preceding one-axis sensitivities into the procurement-relevant capital-cost test. Both axes sweep simultaneously: SMR capital cost runs from NOAK to FOAK on the vertical axis (\$2,250–14,700 kW_e⁻¹), and absorption-chiller capital cost runs from Bare-Low to Turnkey-High on the horizontal axis (\$450–1,200 kW_c⁻¹). Every cell in the 5 × 5 grid is solved at 2023 ATB-Mid market conditions, full-load PUE 1.35, WACC 6.7%, and zero carbon price, with the production tax credit applied. A viable region now opens along the NOAK edge:

every cell in the NOAK row reaches a positive grid-cost margin, from +27% with Bare-Low absorption to +14% with Turnkey-High, so the margin = 0 contour enters the published rectangle. The most favorable corner (NOAK reactor with a Bare-Low absorption chiller at $\$450 \text{ kW}_c^{-1}$) lands at +27%, and the least favorable (FOAK reactor with Turnkey-High absorption) at -383%. The frontier is governed almost entirely by the reactor axis: absorption capital shifts the margin by at most 13 percentage points across its full published range, whereas raising reactor capital one trajectory above NOAK drops the margin below -60%. Reaching roughly nth-of-a-kind reactor capital is therefore the binding condition. With the production tax credit it makes colocated SMR with absorption cooling cheaper than grid supply at 2023 ERCOT prices and zero carbon cost, and no tested capital trajectory above the NOAK row is competitive.

2.5 Carbon pricing is a contingent second route

A carbon price provides a weaker, second route to nuclear competitiveness, distinct from the NOAK capital route just described. Holding capital at the ATB-Mid baseline, the policy axis is the carbon price (Figure 5a). All four cases have net annual CO_2 footprints that are nearly fixed by dispatch (Case 0 +369, Case 3 +486, Case 1 -388 and Case 2 -448 kt $\text{CO}_2 \text{ yr}^{-1}$), so each TAC-carbon relationship is essentially linear over the plotted range. The nuclear footprints barely deepen as carbon is valued, because the production tax credit already drives the reactor to export close to its full surplus, leaving little additional export to unlock.

The grid-only Case 0 increases by $\$0.37 \text{ M yr}^{-1}$ for each $\$1 \text{ tCO}_2^{-1}$ rise in carbon price because the ERCOT grid mix is carbon-intensive. The two nuclear cases decrease by about $\$0.39\text{--}0.45 \text{ M yr}^{-1}$ under the same price increment, because their grid exports displace carbon-intensive generation and receive a corresponding credit. Case 3 NGCC increases by $\$0.49 \text{ M yr}^{-1}$ for each $\$1 \text{ tCO}_2^{-1}$ rise, carrying the full direct emissions on the buyer side. As the carbon price rises, the two on-site options cross the grid-only baseline in opposite directions. At ATB-Mid capital the nuclear cases start well above the baseline and fall below it between $\$142$ and $\$151 \text{ tCO}_2^{-1}$, while NGCC starts below grid and rises above it near $\$191 \text{ tCO}_2^{-1}$; both crossings are extrapolations beyond the $\$0\text{--}\100 tCO_2^{-1} range solved in group G6. Above about $\$151 \text{ tCO}_2^{-1}$ both nuclear cases are cheaper than grid supply at mid-range capital, and by the NGCC-grid crossover near $\$191 \text{ tCO}_2^{-1}$ they are below both grid procurement and the on-site gas alternative. This carbon route is independent of the NOAK capital route: at NOAK capital the nuclear cases already sit below grid cost at zero carbon price.

The cogeneration cooling earns only a small, indirect carbon benefit: meeting cooling from steam rather than electricity lets Case 2 both export more and import less than Case 1, giving it a marginally deeper net footprint (-448 versus -388 kt $\text{CO}_2 \text{ yr}^{-1}$) and a slightly earlier crossover, but it does not create a separate credit mechanism. Both crossings depend on crediting the reactor's merchant exports at the ERCOT hourly average carbon intensity: the nuclear cases reach a net-negative footprint only because the $\sim 1.43 \text{ TWh yr}^{-1}$ of exported electricity displaces grid generation. The carbon route is correspondingly sensitive to this accounting choice. Halving the export credit roughly doubles the crossover price, and removing it entirely

260 pushes the crossover far above any carbon price modeled here, leaving NOAK reactor
 261 capital with the production tax credit as the route that actually makes the nuclear
 262 cases competitive.

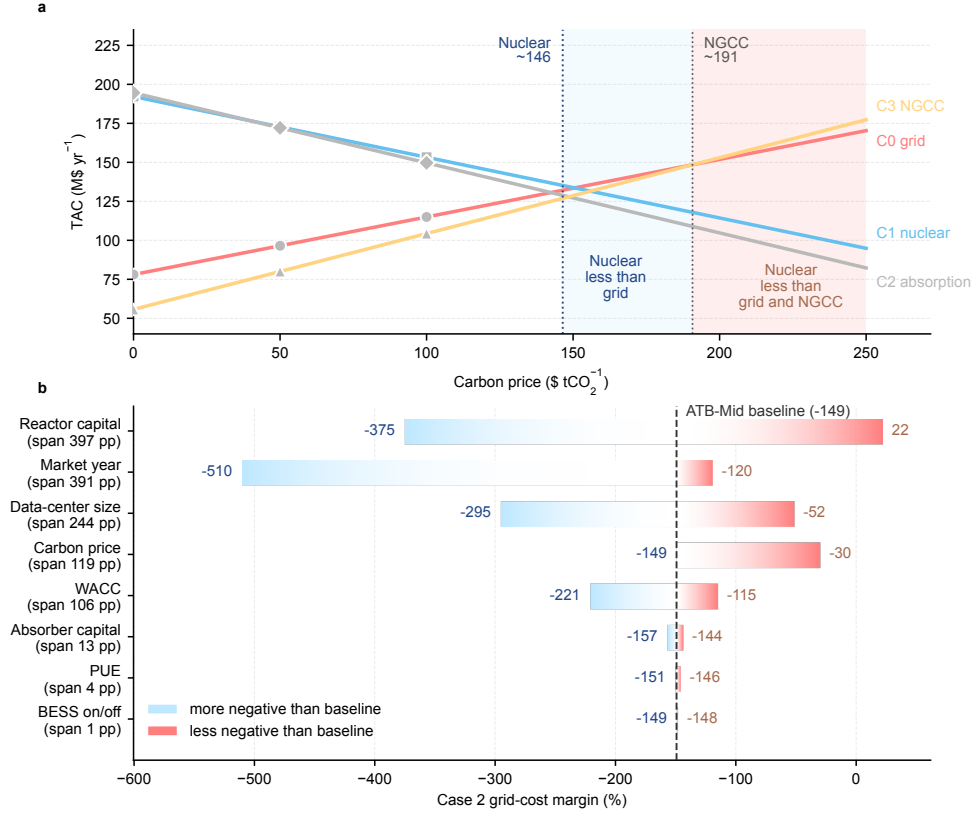


Fig. 5 Policy crossover and global sensitivity ranking. TAC denotes total annualized cost; NGCC, natural gas combined cycle; ATB-Mid, mid-range National Laboratory of the Rockies Annual Technology Baseline reactor-capital case; SMR, small modular reactor; BESS, battery energy storage system; PUE, power usage effectiveness; WACC, weighted-average cost of capital; CAPEX, overnight capital expenditure. (a) TAC extrapolated against carbon price for the four configurations. Vertical dotted lines mark the nuclear and NGCC crossovers against Case 0; the blue region marks where the nuclear cases begin to beat grid supply, and the pink region marks the higher-carbon-price range where they are below both grid and NGCC. (b) Ranked sensitivity of the Case 2 grid-cost margin to each boundary axis, anchored at the 2023 ATB-Mid baseline (−149%).

263 3 Discussion and conclusions

264 Colocated small modular reactors with absorption cooling are a conditional, not cat-
 265 egorical, low-carbon strategy for hyperscale data centers. The condition is narrower
 266 than the usual heat-recovery argument suggests: across the four configurations and

100 scenarios, competitiveness is governed mainly by reactor capital cost interacting with the Section 45U production tax credit. At 2023 mid-range capital, the reactor-with-absorption configuration costs 149% more than grid supply even after the credit. At nth-of-a-kind capital, the same policy support makes it 22% cheaper than grid supply with no carbon price. The central conclusion is therefore that nuclear data-center cogeneration becomes viable only when reactor capital reaches the low-cost learning trajectory; heat recovery does not set the boundary.

The mechanism is a fixed-cost threshold rather than an operating-efficiency threshold. Sweeping reactor and absorption-chiller capital together, the entire nth-of-a-kind reactor row is competitive, from +27% with a bare-low absorption chiller to +14% with a turnkey-high one, whereas one reactor-capital trajectory above nth-of-a-kind pushes the margin below -60%. Absorption capital changes the margin by at most 13 percentage points over its published range. By contrast, the reactor-capital trajectory moves the margin by hundreds of percentage points, so the production tax credit is decisive only once the pre-credit capital gap is already small enough to close.

This finding changes how the data-center nuclear question should be framed. Unit-size matching removes an important distortion but does not create parity: scaling the load to $2.5\times$ brings the reactor close to import-export balance, yet the cogeneration plant remains 62% more expensive than the matched grid baseline. Market year mainly determines whether the on-site gas comparator is temporarily attractive; it does not rescue mid-range nuclear capital. Financing helps by lowering annualized fixed cost, but the 5–10% WACC sweep still leaves Case 2 far above grid supply at ATB-Mid capital. These results suggest that procurement discussions should treat load matching, market regime and financing as necessary supporting conditions rather than substitutes for reactor cost learning.

A carbon price is a second route but a less robust one. Holding reactor capital at the mid-range baseline, the nuclear cases cross the grid baseline only near $\$150 \text{ tCO}_2^{-1}$, and this crossing depends on crediting merchant nuclear exports at the ERCOT hourly average carbon intensity. Halving the export credit roughly doubles the crossover price, and removing it pushes the crossover beyond the modelled policy range. The low-capital route is therefore more decision-relevant than the carbon-price route because it does not depend on a particular avoided-emissions accounting convention.

Against this capital-dominated boundary, cooling integration should be interpreted as a reliability and energy-management design choice rather than the main economic source of value. The absorption chiller supplies $862 \text{ GWh}_c \text{ yr}^{-1}$, about 94% of annual cooling, but it raises cost by $\$2.4 \text{ M yr}^{-1}$ at the baseline because its capital cost, the lost turbine output and a smaller production tax credit slightly exceed the avoided electric cooling cost (Section 2.3). The sign reverses when full-load PUE exceeds about 1.40 or when carbon is priced, so absorption can matter for inefficient campuses or carbon-constrained settings. For an efficient COP-3.2 electric chiller, however, thermal integration is near cost-neutral. This refines waste-heat studies that treat absorption as categorically beneficial and reactor techno-economic studies that omit cooling-side physics.

Several limitations bound the quantitative margins without overturning the central ordering. The load-scaling sweep uses one hyperscale profile, so campuses dominated

by different artificial-intelligence workloads or located in different climates could shift the cooling value. The carbon-policy route uses hourly average rather than marginal emission factors, so a merit-order marginal series would better estimate displaced emissions from nuclear exports. Demand charges and ancillary-service revenues are not modelled; adding delivered retail tariffs or market-service revenues would change the grid and export economics. The gas comparator omits forced outages, redundant generation and an outage-covering grid tie, so five-nines availability requirements would erode its baseline advantage. The production tax credit is applied at full prevailing-wage value to net nuclear generation under an assumed power-purchase-agreement structure and held at nominal in-window values, so eligibility limits, credit expiry or inflation-indexing details would alter the low-capital result. The reactor is also treated as the smallest deployable commercial unit rather than an optimized multi-module buildout.

For procurement, the result is a short list of conditions that must coincide: reactor capital on the nth-of-a-kind trajectory, production-tax-credit eligibility, a load large enough to use the smallest commercial reactor and a market regime in which grid procurement is not unusually cheap. Where those conditions hold, small modular reactor cogeneration can undercut grid supply without a carbon price, and absorption cooling can be chosen for operational reasons rather than as the source of the saving. Where reactor capital remains at first-of-a-kind or mid-range levels, neither thermal integration nor a plausible carbon price is enough. The broader contribution of the study is the boundary map itself: it provides a way to test nuclear data-center proposals against region-specific tariffs, financing, cooling climates and carbon-accounting rules before treating firm low-carbon supply as an economic proposition.

4 Methods

4.1 System configurations

The analysis compares four configurations that supply electricity and chilled water to a single 200 MW_e rated hyperscale data center in Houston, Texas. Each configuration uses the same data-center load profile, vapor-compression chiller technology when required, and financial parameters; the configurations differ only in the electricity supply pathway and in whether the cooling chain uses recovered reactor heat. Figure 1 illustrates the cogeneration mechanism and the grid and storage interfaces for the thermal-integration case. Cases 0, 1 and 3 are reduced versions of this layout in which selected blocks are disabled. Table 1 summarizes the four configurations.

Case 0 represents the conventional grid-supplied data center. All electricity is purchased from ERCOT at the day-ahead locational marginal price, and all cooling is produced by a water-cooled centrifugal vapor-compression chiller with a Houston-derated coefficient of performance of 3.2. This case serves as the benchmark for the grid-cost margin reported throughout the paper. Case 1 replaces grid imports with on-site BWRX-300 generation while retaining the same VCC. Heat at the turbine condenser is rejected to the cooling tower and is not recovered. Bidirectional grid exchange is retained so that excess electricity can be sold to ERCOT during low-load hours and grid imports can cover residual demand during planned outages. Case 2

Table 1 System configurations modeled in this study. ✓ denotes that the block is enabled; – that it is disabled. SMR denotes small modular reactor; NGCC, natural gas combined cycle; ERCOT, Electric Reliability Council of Texas; PCC, point of common coupling; VCC, vapor-compression chiller; LiBr, lithium bromide; BESS, battery energy storage system.

		Case 0	Case 1	Case 2	Case 3
Generation	BWRX-300 SMR + steam turbine	–	✓	✓	–
	NGCC on-site, F-class	–	–	–	✓
	ERCOT grid (PCC, 300 MW)	✓ (import)	✓ (bi-dir.)	✓ (bi-dir.)	–
Cooling	VCC, water-cooled	✓ (primary)	✓ (primary)	✓ (backup)	✓ (primary)
	Double-effect LiBr absorber	–	–	✓	–
Storage	Li-ion BESS, 100 MWh, 50 MW	–	sensitivity	sensitivity	–

adds a double-effect LiBr–H₂O absorption chiller fed by a mid-pressure extraction tap on the high-pressure (HP) turbine. The cascade keeps the low-pressure (LP) turbine in service, so diverted steam incurs a Willans penalty rather than the full turbine efficiency penalty. A 160 MW_e vapor-compression backup chiller remains installed, the same capacity as the primary chiller in the VCC-only cases, because it must carry the full cooling load whenever the absorption unit shuts down at the LiBr crystallization threshold. Case 3 replaces the SMR with a 200 MW_e on-site F-class natural gas combined cycle, retains the VCC, and disconnects from the grid to isolate the value of replacing grid procurement with a fossil-fueled on-site alternative.

The data-center IT load follows an annual hourly trace synthesized from a 200 MW_e hyperscale reference profile with a peak hour of 142.4 MW_e, an annual average of 94.7 MW_e, and an annual utilization of 0.67. The same trace is used in every case and every scenario; the IT load is not a decision variable. We treat the full IT electrical draw as heat that the cooling system must reject, so the hourly cooling demand is

$$Q_{\text{cool}}^{\text{dem}}(t) = \frac{P_{\text{IT}}(t)}{\eta_{\text{ch}}}, \quad (1)$$

where $\eta_{\text{ch}} = 0.90$ accounts for the chilled-water distribution chain. Each scenario is labeled by its full-load vapor-compression PUE, $1 + 1/(\eta_{\text{ch}} \text{COP}_v)$, which is 1.35 at the baseline COP of 3.2. The dispatch-realized annual PUE, $1 + P_{\text{cool}}/P_{\text{IT}}$, is slightly lower for the vapor-compression cases, near 1.33, because the part-load COP curve runs more efficiently than its full-load point. It falls to about 1.05 in Case 2, where steam-driven absorption supplies most of the cooling and leaves little electric cooling load. The cooling-efficiency sensitivity in Section 2.3 sweeps the chiller COP so that this full-load PUE label spans {1.10, 1.30, 1.50}, the range between high-efficiency and legacy hyperscale practice^{7,8}.

4.2 Optimization formulation

The four cases are formulated as separate dispatch problems on the same 8760-hour horizon. Cases 0 and 3 reduce to closed-form annual cost evaluations because the absence of storage and heat recovery makes the hourly decisions independent. Cases 1 and 2 are mixed-integer linear programs: the only integer structure is the special-ordered-set (SOS2) representation of the vapor-compression chiller part-load curve described below, which is non-convex and cannot be relaxed; no binary commitment variables are introduced, because round-trip losses in the battery block already make simultaneous charge and discharge unprofitable. The same accounting structure is applied to all four cases so that cross-case comparisons remain on a common basis.

Reactor and turbine block (Cases 1 and 2). The BWRX-300 reactor is modeled as a continuous thermal source with bounded ramping and an annual capacity-factor constraint. The reactor block enforces

$$\underline{P}_{\text{rx}} \leq P_{\text{rx}}(t) \leq \overline{P}_{\text{rx}}, \quad \forall t \in \mathcal{T}, \quad (2)$$

$$|P_{\text{rx}}(t) - P_{\text{rx}}(t-1)| \leq \Delta P_{\text{rx}}^{\text{max}}, \quad \forall t \geq 1, \quad (3)$$

$$\sum_{t \in \mathcal{T}} P_{\text{rx}}(t) \geq \overline{\text{CF}} \cdot \overline{P}_{\text{rx}} \cdot |\mathcal{T}|, \quad (4)$$

with $\overline{P}_{\text{rx}} = 870 \text{ MW}_{\text{th}}$, $\underline{P}_{\text{rx}} = 435 \text{ MW}_{\text{th}}$ (50% minimum-load fraction from the BWRX-300 design basis²¹), $\Delta P_{\text{rx}}^{\text{max}} = 522 \text{ MW}_{\text{th}} \text{ h}^{-1}$ corresponding to a 1 % min^{-1} ramp, and $\overline{\text{CF}} = 0.92$ from the mature U.S. light-water-reactor fleet average²². The annual capacity-factor constraint sets an annual energy floor equivalent to roughly one month of downtime, without modeling the timing of individual outages or introducing unit-commitment binaries.

The Rankine steam cycle is represented by a cascaded high-pressure to low-pressure turbine with a Willans-line extraction penalty. Letting $\eta_r = 0.345$ be the rated electric efficiency at zero extraction and $\alpha_w = 0.083 \text{ MW}_e \text{ MW}_{\text{th}}^{-1}$ the marginal electricity loss intensity for mid-pressure steam diverted to the absorber:

$$P_{\text{tg}}(t) = \eta_r P_{\text{rx}}(t) - \alpha_w Q_{\text{ext}}(t), \quad (5)$$

$$P_{\text{tn}}(t) = (1 - \xi_{\text{aux}}) P_{\text{tg}}(t), \quad (6)$$

with $\xi_{\text{aux}} = 0.10$. Because diverted steam still expands through the high-pressure turbine before extraction, the Willans coefficient is roughly one order of magnitude smaller than charging the full turbine efficiency against extracted heat, as would be appropriate for a parallel main-steam tap.

Absorption-chiller submodel (Case 2). The double-effect LiBr–H₂O absorption chiller is described by a time-varying coefficient of performance and a binary availability indicator that captures LiBr crystallization risk^{18,19}. With $T_{\text{wb},0} = 26^\circ\text{C}$ as the Houston design wet-bulb, $\text{COP}_a^0 = 1.10$ as the COP at the design wet-bulb, $\delta = 0.015^\circ\text{C}^{-1}$ as the linear de-rating slope, and $\overline{\text{COP}}_a = 1.30$ as the nameplate cap,

410 the chiller COP at wet-bulb $T_{\text{wb}}(t)$ is

$$\text{COP}_a(T_{\text{wb}}(t)) = \max \left\{ \min [\text{COP}_a^0 - \delta (T_{\text{wb}}(t) - T_{\text{wb},0}), \overline{\text{COP}}_a], 0.5 \right\}. \quad (7)$$

411 The lower bound of 0.5 is a numerical safeguard; Houston wet-bulbs in the weather
 412 record do not reduce a physical chiller to this value because the crystallization gate
 413 engages first. With $\Delta T_{\text{cw}} = 5$ K as the cooling-tower approach, $T_{\text{cry}} = 32^\circ\text{C}$ as the
 414 LiBr-crystallization-line cooling-water inlet, and $a_m = 0.95$ a maintenance availability
 415 that derates the unit to account for scheduled servicing, the availability indicator is

$$a(t) = \begin{cases} a_m & \text{if } T_{\text{wb}}(t) + \Delta T_{\text{cw}} \leq T_{\text{cry}}, \\ 0 & \text{otherwise,} \end{cases} \quad (8)$$

416 so the absorption chiller is forced fully offline whenever the cooling-water inlet would
 417 exceed the LiBr crystallization line, and otherwise operates at its maintenance-derated
 418 availability. Equivalently, with the $\Delta T_{\text{cw}} = 5$ K tower approach this is a wet-bulb gate
 419 at $T_{\text{cry}} - \Delta T_{\text{cw}} = 32 - 5 = 27^\circ\text{C}$, the value marked in Figure S1a. Delivered absorption
 420 cooling is

$$Q_a(t) = \text{COP}_a(T_{\text{wb}}(t)) \cdot a(t) \cdot Q_{\text{ext}}(t), \quad (9)$$

421 and a parasitic electric load $\beta_a Q_a(t)$ with $\beta_a = 0.020 \text{ kW}_e \text{ kW}_c^{-1}$ powers solution and
 422 refrigerant pumps. The vapor-compression backup carries the residual cooling load
 423 whenever $a(t) = 0$.

424 **Cooling, BESS and grid blocks.** Vapor-compression cooling links its electric
 425 input to its cooling output through a part-load coefficient of performance. A water-
 426 cooled centrifugal chiller gains efficiency at reduced load, with the coefficient of
 427 performance peaking near 40–50% of rated duty before declining at very low load, fol-
 428 lowing the integrated part-load value of ANSI/AHRI Standard 550/590²³. The chiller
 429 electricity is therefore a non-convex, piecewise-linear function of the cooling delivered,
 430 $P_v(t) = f_v(Q_v(t))$, anchored at the Houston-derated full-load value $\text{COP}_v = 3.2$. In
 431 the closed-form Cases 0 and 3 the chiller is evaluated directly at the realized load
 432 fraction; in the nuclear Cases 1 and 2 the same curve enters the optimization as an
 433 SOS2 piecewise map, the single source of integer structure in the model. The lithium-
 434 ion BESS, sized at 100 MWh and 50 MW (C/2), follows the standard state-of-charge
 435 dynamics

$$E_B(t) = (1 - \rho_B) E_B(t-1) + (\sqrt{\eta_{\text{rt}}} B^+(t) - B^-(t)/\sqrt{\eta_{\text{rt}}}) \Delta t, \quad (10)$$

436 with cyclical closure $E_B(|\mathcal{T}|) = E_B(0)$, $\eta_{\text{rt}} = 0.85$, $\rho_B = 4.17 \times 10^{-5} \text{ h}^{-1}$, and state-of-
 437 charge bounds $[0.05, 0.95] \text{ Cap}_E$. Grid exchange is bounded by the point of common
 438 coupling, $0 \leq P_g^+(t)$, $P_g^-(t) \leq \bar{P}_{\text{pcc}}$, with $\bar{P}_{\text{pcc}} = 300$ MW for the baseline campus; in
 439 the size-matching sweep the interconnect scales with the data-center load, since it is
 440 part of the data-center build rather than the fixed reactor.

441 **Energy balances.** Two equality constraints hold every hour. The electric balance
 442 equates supply, BESS discharge and grid imports to the data-center IT load, chiller
 443 drives, absorber parasitics, BESS charge and grid exports:

$$P_{\text{tn}}(t) + P_{\text{g}}^+(t) + B^-(t) = P_{\text{IT}}(t) + P_{\text{v}}(t) + \beta_{\text{a}}Q_{\text{a}}(t) + P_{\text{g}}^-(t) + B^+(t). \quad (11)$$

444 The cooling balance equates total cooling supplied to the cooling demand of Eq. (1):

$$Q_{\text{a}}(t) + Q_{\text{v}}(t) = Q_{\text{cool}}^{\text{dem}}(t). \quad (12)$$

445 For Cases 0, 1 and 3 the absorption variables $Q_{\text{a}}(t)$, $Q_{\text{ext}}(t)$ are fixed to zero; for
 446 Case 3 the reactor and turbine variables are removed and the IT and chiller load is
 447 served entirely by the NGCC block at a full-load higher-heating-value (HHV) efficiency
 448 $\eta_{\text{hhv}} = 0.495$, with the part-load heat-rate penalty of Eq. (18).

449 **Objective.** The optimization minimizes TAC summed over seven accounting
 450 buckets:

$$\begin{aligned} \min \text{ TAC} = & \sum_k \text{CRF}_k \text{CAPEX}_k S_k + \text{FOM} + \text{VOM} \\ & + \Phi_{\text{fuel}} + \Phi_{\text{grid}} + \Phi_{\text{CO}_2} - \Phi_{\text{ptc}}, \end{aligned} \quad (13)$$

451 where the first term annualizes overnight capital expenditure (CAPEX) across reac-
 452 tor, turbine, absorption and VCC chillers, BESS and NGCC sized at S_k , each
 453 amortized over its own engineering life N_k through an asset-specific capital recov-
 454 ery factor (CRF; reactor and turbine 20 yr, chillers 25 yr, NGCC 30 yr, battery
 455 15 yr); $\text{FOM} = \sum_k \text{FOM}_k S_k$ aggregates fixed operations and maintenance (FOM),
 456 and v_{x} denotes the variable operations and maintenance (VOM) coefficient of stream
 457 x (Table S3); the variable terms are

$$\begin{aligned} \text{VOM} = & \sum_t [v_{\text{rx}}P_{\text{tn}}(t) + v_{\text{v}}Q_{\text{v}}(t) + v_{\text{a}}Q_{\text{a}}(t)]\Delta t \\ & + \sum_t v_{\text{B}}(B^+(t) + B^-(t))\Delta t, \end{aligned} \quad (14)$$

$$\Phi_{\text{fuel}} = \pi_{\text{f}} \sum_t P_{\text{rx}}(t)\Delta t, \quad (15)$$

$$\Phi_{\text{grid}} = \sum_t \pi_{\text{g}}(t)[P_{\text{g}}^+(t) - P_{\text{g}}^-(t)]\Delta t, \quad (16)$$

$$\Phi_{\text{CO}_2} = \pi_{\text{CO}_2} \Theta \sum_t [\gamma_{\text{rx}}P_{\text{tn}}(t) + \theta(t)(P_{\text{g}}^+(t) - P_{\text{g}}^-(t))]\Delta t. \quad (17)$$

458 For Case 3, the same accounting buckets are used but the reactor terms are replaced
 459 by the NGCC terms. The combined-cycle heat rate rises below the design point, so
 460 fuel use and combustion emissions are both scaled by a part-load multiplier $\phi_{\text{ng}}(\ell) \geq$
 461 1 of the electrical load fraction $\ell(t) = P_{\text{ng}}(t)/\bar{P}_{\text{ng}}$, taken from a representative F-
 462 class combined-cycle characteristic and equal to one at full load. The VOM term uses

463 $v_{\text{ng}}P_{\text{ng}}(t) + v_v Q_v(t)$, where $P_{\text{ng}}(t)$ is the NGCC electric output required by the IT and
 464 VCC loads, and the fuel and carbon terms become

$$\Phi_{\text{fuel}}^{\text{ng}} = \sum_t \pi_{\text{ng}}(t) h_{\text{ng}} \phi_{\text{ng}}(\ell(t)) P_{\text{ng}}(t) \Delta t, \quad (18)$$

$$\Phi_{\text{CO}_2}^{\text{ng}} = \pi_{\text{CO}_2} \Theta \sum_t \gamma_{\text{ng}} \phi_{\text{ng}}(\ell(t)) P_{\text{ng}}(t) \Delta t, \quad (19)$$

465 with full-load heat rate $h_{\text{ng}} = 3.412/\eta_{\text{hhv}}$ MMBtu MWh_e⁻¹ and full-load lifecycle factor
 466 $\gamma_{\text{ng}} = 420$ g CO₂-eq kWh_e⁻¹, equal to 360 direct-combustion and 60 upstream-methane
 467 g CO₂-eq kWh_e⁻¹²⁴. Applying the same ϕ_{ng} to both terms holds the fuel carbon con-
 468 tent constant. The carbon term Φ_{CO_2} multiplies the case carbon footprint by a uniform
 469 price π_{CO_2} and a unit-conversion factor $\Theta = 10^{-3}$ t CO₂ kg⁻¹. Exports credit the case
 470 at the ERCOT hourly average carbon intensity $\theta(t)$, derived from the U.S. Energy
 471 Information Administration (EIA)-930 generation mix²⁵, to reflect the time-varying
 472 emissions of the grid power exchanged, following time-resolved hourly emissions
 473 accounting rather than a static annual-average factor²⁶; this hourly average emission
 474 factor is used as a tractable proxy for the true marginal emission factor^{27,28}. Each asset
 475 is annualized over its own engineering life through $\text{CRF}_k = i(1+i)^{N_k}/[(1+i)^{N_k} - 1]$,
 476 with asset life N_k as listed above and baseline WACC $i = 6.7\%$; the battery addi-
 477 tionally carries a 5% end-of-life salvage credit. The WACC sensitivity in Section 2.4
 478 sweeps i over $\{5\%, 6.7\%, 10\%\}$.

479 **Nuclear production tax credit.** The nuclear cases earn the zero-emission nuclear
 480 production tax credit (Section 45U; 26 U.S.C. § 45U), enacted in the Inflation Reduc-
 481 tion Act of 2022²⁹, on their generation^{30,31}. At the prevailing-wage rate the credit is
 482 \$15 MWh_e⁻¹ (1.5¢ kWh_e⁻¹) and is subject to the statutory gross-receipts phaseout: it
 483 holds at the full value while the hourly market price π_g stays at or below the threshold
 484 $\pi_1 = \$25$ MWh_e⁻¹ (2.5¢ kWh_e⁻¹), then declines by \$0.80 for each \$1 MWh_e⁻¹ above it,
 485 reaching zero at $\pi_2 = \$43.75$ MWh_e⁻¹ (4.375¢ kWh_e⁻¹),

$$c_{45\text{U}}(\pi) = \begin{cases} \bar{c}, & \pi \leq \pi_1, \\ \bar{c} \left(1 - \frac{\pi - \pi_1}{\pi_2 - \pi_1}\right), & \pi_1 < \pi < \pi_2, \\ 0, & \pi \geq \pi_2, \end{cases} \quad (20)$$

486 with full credit $\bar{c} = \$15$ MWh_e⁻¹. The annual credit is applied to net nuclear generation,

$$\Phi_{\text{ptc}} = \sum_t c_{45\text{U}}(\pi_g(t)) P_{\text{tn}}(t) \Delta t, \quad (21)$$

487 using the hourly ERCOT price as the kWh-indexed gross-receipts proxy. The slope and
 488 breakpoints reproduce the statutory 16% reduction at the prevailing-wage multiplier.
 489 We treat the credit as a baseline policy condition, assuming a power-purchase-
 490 agreement structure in which the data center is an unrelated offtaker so that delivered
 491 nuclear generation qualifies; statutory amounts are inflation-indexed after 2024 and

the program runs through tax year 2032, both of which we hold at their nominal in-window values. Cases 0 and 3 carry no nuclear generation and therefore no credit.

The cross-case figure of merit used throughout the rest of the paper is the grid-cost margin

$$M_c = 1 - \frac{\text{TAC}_c}{\text{TAC}_0} \Big|_{\text{matched scenario}}, \quad (22)$$

defined as one minus the ratio of the case TAC to the Case 0 (grid-only) TAC at the same year, PUE and carbon price. A positive margin means the configuration is cheaper than grid procurement and a negative margin means it is more expensive; the magnitude is the relative cost difference. We report it for all four configurations as a common measure of cost competitiveness against the grid.

4.3 Data inputs

Inputs to the optimization fall into three families: exogenous time series (hourly traces of IT load, electricity price, average carbon intensity and wet-bulb temperature), equipment parameters (capital and operating cost coefficients plus technical bounds for each plant block), and financial parameters (project lifetime and weighted-average cost of capital). All runs use 8760-hour annual operating horizons; leap-year raw files are trimmed to 8760 hourly records for cross-year comparability. The four characteristic input traces are shown in Supplementary Figure S1; aggregated values and source citations are listed in Supplementary Tables S2 and S3.

The Houston wet-bulb temperature distribution (Supplementary Figure S1a) drives the absorption-chiller availability constraint in Equation (8). Monthly box plots from the Open-Meteo historical-weather archive³² for the 2023 calendar year show that wet-bulb exceeds 26 °C (the absorption design anchor $T_{\text{wb},0}$) in about 32% of summer hours and crosses the LiBr crystallization gate at 27 °C in about 5% of summer hours, concentrated in July and August afternoons. The ERCOT day-ahead LMP duration curves (Supplementary Figure S1b), retrieved through gridstatus.io³³, capture the recent range of grid-procurement cost: 2022 represents the high natural-gas price regime, with an annual mean LMP of \$70 MWh_e⁻¹ and frequent excursions above \$200 MWh_e⁻¹; 2023 was a moderate year at \$57 MWh_e⁻¹; and 2024 declined to \$28 MWh_e⁻¹ as gas prices receded and renewable generation added supply. The IT-load profile (Supplementary Figure S1c) follows a typical hyperscale shape with mild diurnal seasonality and an annual capacity factor of 0.67; peak load reaches 142.4 MW_e and annual mean load is 94.7 MW_e. The absorption-chiller COP_a(T_{wb}) curve (Supplementary Figure S1d) parameterizes Equation (7): COP decreases linearly from the nameplate cap of 1.30 at cool ambient wet-bulb to 1.10 at the design anchor of 26 °C, after which the crystallization gate forces a binary cutoff and the VCC backup carries the cooling load.

Scalar inputs are summarized in Supplementary Table S2, and the per-asset cost and technical parameters carried through the CAPEX, FOM, and VOM terms in Equation (13) are listed in Supplementary Table S3. Reactor capital cost trajectories follow the NLR 2024 Annual Technology Baseline¹¹ at \$14,700, \$7,615 and \$2,250 kW_e⁻¹ for FOAK, Moderate and NOAK cases. The absorption-chiller capital cost grid is anchored at \$750 kW_c⁻¹ baseline and brackets the literature range from

534 \$450 kW_c⁻¹ (Bare-Low, Chinese manufacturer ex-works) to \$1,200 kW_c⁻¹ (Turnkey-
535 High, Western turnkey engineering, procurement and construction). The NGCC cost
536 block follows the NLR ATB 2024 F-class Moderate values; natural-gas fuel cost is
537 the hourly Henry Hub spot price³⁴ plus the Houston Ship Channel basis (mean
538 \$0.10 MMBtu⁻¹). The data audit trail and provenance summary are provided in the
539 Supplementary Information.

540 4.4 Scenario design

541 Eight sensitivity groups are evaluated relative to the baseline 2023, full-load PUE 1.35,
542 ATB-Mid CAPEX and WACC 6.7% configuration to probe the boundary conditions
543 of the cogeneration economics. The full run ledger, consisting of four baseline case
544 runs plus 96 sensitivity runs, is given in Supplementary Table S7.

545 The cooling-efficiency sweep tests the sensitivity of the absorption value proposition
546 to the electric cooling efficiency, implemented by changing the VCC full-load COP and
547 reporting the equivalent full-load PUE while holding the physical heat-rejection load
548 fixed at Eq. (1). The year sweep covers the recent range of Henry Hub and ERCOT
549 prices. The BESS sweep tests whether 100 MWh of storage adds value at current
550 Li-ion capital costs in the nuclear cases, with Case 0 and Case 3 retained as no-
551 storage control rows. The SMR CAPEX sweep covers the published FOAK-to-NOAK
552 range. The two-dimensional SMR $\times$ absorption CAPEX grid (group G5) is the central
553 scenario: it asks whether any joint capital-cost trajectory makes Case 2 cheaper than
554 Case 0 at the 2023 baseline. The carbon-price sweep identifies the carbon price at
555 which the nuclear cases cross the grid-only baseline. The WACC sweep isolates the
556 financing effect on Case 2 with the operating parameters held constant. The data-
557 center size sweep (group G8) scales the IT-load trace and data-center-side cooling and
558 NGCC capacities from 0.5 $\times$ to 3.0 $\times$ while holding the commercial BWRX-300 unit
559 fixed, isolating the effect of unit-size matching.

560 Data availability

561 All data that support the findings of this study, including the input parameters, sce-
562 nario definitions, run-level summaries, hourly dispatch traces and figure source data,
563 are available in the project GitHub repository at <https://github.com/Henryutd2021/Nuclear-DC>
564 and will be deposited in a persistent public archive (Zenodo) with a
565 citable DOI on acceptance. The individual exogenous data sources and their prove-
566 nance are listed in Supplementary Note 4. No proprietary or restricted-access data are
567 used.

568 Code availability

569 The optimization framework, the scenario driver and the figure-generation notebooks
570 that reproduce every result reported here are available in the same GitHub repository
571 and will be archived alongside the data on acceptance.

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573 Author contributions

574 H.L. conceived the study, developed the optimization framework and the hourly
575 absorption submodel, curated the input data, ran the scenarios, produced the figures
576 and wrote the original draft. B.S. contributed to the methodology, validated the
577 optimization results, and reviewed and edited the manuscript. J.Z. conceived and
578 supervised the study, provided resources and funding, and reviewed and edited the
579 manuscript. All authors approved the final manuscript.

580 Competing interests

581 The authors declare no competing interests.

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Supplementary Information

Supplementary Note 1: Nomenclature

The complete set of model sets, decision variables, parameters and abbreviations is listed in Supplementary Table S1.

Table S1: Nomenclature for the optimization model: sets and indices, decision variables, parameters and abbreviations.

Symbol	Description
Sets and indices	
$t \in \mathcal{T}$	hourly time step over an annual modeling horizon, $ \mathcal{T} = 8760$
$c \in \mathcal{C}$	system configuration, $\mathcal{C} = \{0, 1, 2, 3\}$
Decision variables (hour t)	
$P_{rx}(t)$	reactor thermal output (MW_{th})
$P_{tg}(t)$	turbine gross electric output (MW_e)
$P_{tn}(t)$	turbine net electric output (MW_e)
$Q_{ext}(t)$	mid-pressure steam extraction to absorber (MW_{th})
$Q_a(t)$	absorption-chiller cooling delivered (MW_c)
$P_v(t)$	vapor-compression chiller (VCC) electric input (MW_e)
$Q_v(t)$	VCC cooling delivered (MW_c)
$P_g^+(t)$	grid import (MW_e)
$P_g^-(t)$	grid export (MW_e)
$P_{ng}(t)$	NGCC electric output, Case 3 (MW_e)
$B^+(t)$	BESS charging power (MW_e)
$B^-(t)$	BESS discharging power (MW_e)
$E_B(t)$	BESS state of charge (MWh)
Parameters	
$P_{IT}(t)$	data-center IT load (MW_e)
$\pi_g(t)$	ERCOT day-ahead LMP ($\$ \text{MWh}_e^{-1}$)
$\theta(t)$	ERCOT hourly average carbon intensity ($\text{g CO}_2 \text{ kWh}_e^{-1}$)
$T_{wb}(t)$	Houston wet-bulb temperature ($^\circ\text{C}$)
η_r	turbine rated electric efficiency at zero extraction ($-$)
α_w	Willans extraction penalty ($\text{MW}_e \text{ MW}_{th}^{-1}$)
ξ_{aux}	turbine auxiliary fraction ($-$)
PUE	power usage effectiveness ($-$)

continued on next page

Supplementary Table S1 (continued)

Symbol	Description
η_{ch}	chilled-water chain efficiency (–)
$\text{COP}_{\text{a}}(T_{\text{wb}})$	absorption chiller COP (–)
$a(t)$	absorption availability (a_{m} operating, 0 crystallization)
a_{m}	absorption maintenance availability (–)
COP_{v}	VCC full-load coefficient of performance (–)
$f_{\text{v}}(\cdot)$	VCC part-load electricity map (MW_{e})
β_{a}	absorber parasitic load intensity ($\text{kW}_{\text{e}} \text{ kW}_{\text{c}}^{-1}$)
η_{rt}	BESS round-trip efficiency (–)
ρ_{B}	BESS hourly self-discharge (–)
Cap_{E}	BESS energy capacity (MWh)
$\overline{\text{CF}}$	reactor annual capacity factor (–)
$\overline{P}_{\text{rx}}, \underline{P}_{\text{rx}}$	reactor maximum and minimum thermal output (MW_{th})
$\Delta P_{\text{rx}}^{\text{max}}$	reactor ramp limit ($\text{MW}_{\text{th}} \text{ h}^{-1}$)
$\overline{P}_{\text{pcc}}$	point-of-common-coupling capacity (MW_{e})
CAPEX_k	overnight capital coefficient of asset k
FOM_k	fixed O&M coefficient of asset k
VOM_k	variable O&M coefficient of asset k
CRF_k	capital recovery factor of asset k (–)
N_k	engineering life of asset k (yr)
π_{f}	delivered nuclear fuel cost ($\$ \text{ MWh}_{\text{th}}^{-1}$)
$\pi_{\text{ng}}(t)$	delivered natural-gas price for NGCC (Henry Hub + basis; $\$ \text{ MMBtu}^{-1}$)
h_{ng}	NGCC full-load HHV heat rate ($\text{MMBtu} \text{ MWh}_{\text{e}}^{-1}$)
η_{hhv}	NGCC full-load HHV net efficiency (–)
$\overline{P}_{\text{ng}}$	NGCC rated electric capacity (MW_{e})
$\phi_{\text{ng}}(\ell)$	NGCC part-load heat-rate multiplier (–)
$\bar{c}, \pi_1, \pi_2$	45U full credit and phaseout breakpoints ($\$ \text{ MWh}_{\text{e}}^{-1}$)
π_{CO_2}	carbon price ($\$ \text{ tCO}_2^{-1}$)
Δt	hourly time step length (h)
γ_{rx}	reactor lifecycle emissions ($\text{g CO}_2 \text{ kWh}_{\text{e}}^{-1}$)
γ_{ng}	NGCC lifecycle emissions ($\text{g CO}_2\text{-eq} \text{ kWh}_{\text{e}}^{-1}$)
Θ	kg-to-tonne unit conversion ($= 10^{-3}$)
Abbreviations	
ATB	National Laboratory of the Rockies Annual Technology Baseline

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Supplementary Table S1 (continued)

Symbol	Description
BWRX-300	GE–Hitachi 870 MW _{th} , 270 MW _e Boiling Water small modular reactor
CAPEX	overnight capital expenditure
CRF	capital recovery factor
COP	coefficient of performance
ERCOT	Electric Reliability Council of Texas
FOAK–NOAK	first- and nth-of-a-kind reactor capital cost
FOM, VOM	fixed and variable operations and maintenance
IT	information-technology load
M	grid-cost margin, $1 - \text{TAC}_c / \text{TAC}_0$ (positive = cheaper than grid)
LiBr–H ₂ O	lithium-bromide–water absorption pair
LMP	locational marginal price
NGCC	natural gas combined cycle
PTC	production tax credit (Inflation Reduction Act Section 45U)
PUE	power usage effectiveness
SMR	small modular reactor
TAC	total annualized cost ($\$ \text{yr}^{-1}$)
VCC	vapor-compression chiller
WACC	weighted-average cost of capital

Supplementary Note 2: Input data and equipment parameters

The scalar input data and the per-asset cost and performance parameters carried through the optimization are listed in Supplementary Tables S2 and S3.

Supplementary Note 3: Input traces

The four characteristic exogenous input traces driving the optimization are shown in Supplementary Figure S1.

Supplementary Note 4: Exogenous data sources and provenance

Each exogenous input is drawn from a public, citable source and trimmed to an 8760-hour annual horizon. ERCOT day-ahead locational marginal prices at the Houston load zone are retrieved through the gridstatus.io public data interface³³. Hourly average grid carbon intensities are derived from the U.S. Energy Information Administration EIA-930 Hourly Electric Grid Monitor generation mix²⁵ weighted by United Nations Economic Commission for Europe 2022 lifecycle emission factors for each technology³⁵. Houston wet-bulb temperature is taken from the Open-Meteo historical-weather reanalysis archive³². Delivered natural gas is the U.S. Energy Information

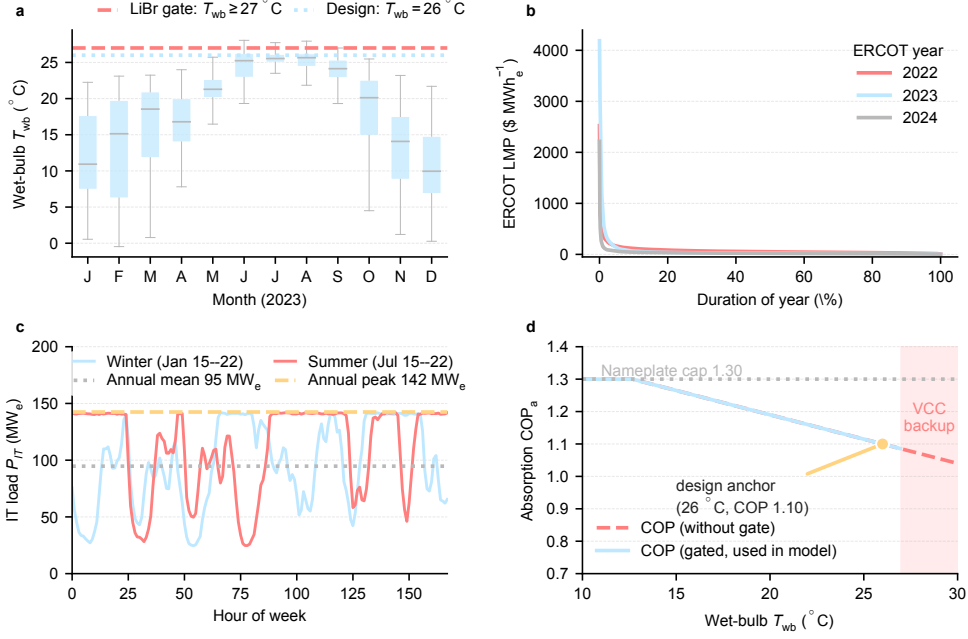


Fig. S1 Hourly input traces driving the optimization. LiBr denotes lithium bromide; COP, coefficient of performance; ERCOT, Electric Reliability Council of Texas; LMP, locational marginal price; IT, information technology; VCC, vapor-compression chiller. (a) Houston monthly wet-bulb distribution from the Open-Meteo historical-weather archive (year 2023). The dashed horizontal line marks the LiBr crystallization gate at $T_{wb} \geq 27^{\circ}$ C; the dotted line marks the COP design anchor $T_{wb,0} = 26^{\circ}$ C. (b) Duration curves of ERCOT day-ahead LMP for the three modeled years (2022 high-gas-price regime; 2023 moderate regime; 2024 low-LMP regime). (c) Data-center IT-load profile for a typical winter and summer week, with annual mean and peak overlaid. (d) Double-effect LiBr-H₂O absorption-chiller COP as a function of wet-bulb temperature; the dashed segment shows the linear de-rating below the crystallization gate, and the shaded band marks the backup-cooling region in which the VCC carries the cooling load.

Administration Henry Hub daily spot price³⁴ plus a fixed Houston Ship Channel basis. Reactor and NGCC overnight capital costs and fixed operations and maintenance follow the National Laboratory of the Rockies 2024 Annual Technology Baseline¹¹; the reactor capacity factor is the U.S. Energy Information Administration fleet average²²; and absorption-chiller capital costs are bracketed from American Society of Heating, Refrigerating and Air-Conditioning Engineers Technical Committee 8.3 absorption-chiller technical data and publicly available vendor literature. The Section 45U production tax credit follows the enacting statute^{29,30} and U.S. Internal Revenue Service guidance³¹. No proprietary or restricted-access data are used, and the audit trail linking each raw file to its processed model input is included in the project repository.

Supplementary Note 5: Market inputs by operating year

The three modeled years bracket the recent range of ERCOT and natural-gas market conditions (Supplementary Table S4). 2022 is the high-gas-price regime, 2023 a moderate year with low gas but firm power prices, and 2024 a low-price year in which grid procurement is comparatively cheap. The on-site natural gas combined cycle (Case 3) beats grid supply only in 2023, when low delivered gas coincides with firm ERCOT prices; in 2022 the gas and grid prices move together and erase the advantage, and in 2024 the low ERCOT price makes grid procurement the cheapest option.

Supplementary Note 6: Cogeneration dispatch and storage detail

Over the full year the absorption chiller delivers 862 GWh_c of cooling and the vapor-compression backup 60 GWh_c. The backup energy is spread over roughly 2,870 hours of partial top-up, which also cover the 5% maintenance derating of the absorber. The crystallization gate of Equation (8) forces the absorber fully offline for about 115 hours, concentrated from late June through August, during which the vapor-compression chiller carries the entire 13.6 GWh_c cooling load of those hours. In the islanded gas case (Case 3) the 2023 day-ahead price falls below the NGCC running cost of about \$21 MWh_e⁻¹ in roughly 42% of hours, but the disconnected plant cannot import in those hours, so its comparison against grid supply is made on annual cost rather than hour by hour. The lithium-ion battery sweep (group G3) lowers total annualized cost by \$0.63 M yr⁻¹ in each nuclear case, an improvement of about 0.8 percentage points in the grid-cost margin that never changes the configuration ranking, so it is reported as a secondary boundary rather than a main result.

Supplementary Note 7: Marginal value of absorption cooling

Supplementary Figure S2 and Supplementary Table S5 report the marginal value of adding the absorption chiller to the SMR-only plant (Case 2 minus Case 1) at the 2023 ATB-Mid baseline and across the cooling-efficiency sweep. The gross avoided vapor-compression electricity grows with the reported PUE, so the chiller moves from a net penalty against an efficient electric chiller (PUE 1.10) to a net saving against an inefficient one (PUE 1.50), with break-even near a full-load PUE of 1.40.

Supplementary Note 8: Data-center size-matching results

Supplementary Figure S3 and Supplementary Table S6 report the size-matching sweep (group G8). Holding the BWRX-300 reactor fixed at 270 MW_e net while scaling the IT load, cooling and grid interconnect from 0.5× to 3.0× moves Case 2 from a deep merchant-export surplus at small scale to a net importer at 3.0×. The annual dollar gap to grid supply is nearly flat because the larger campus also raises the grid-only counterfactual; what improves is the IT-energy-normalized cost intensity, which falls from \$235 to \$143 MWh_{IT}⁻¹.

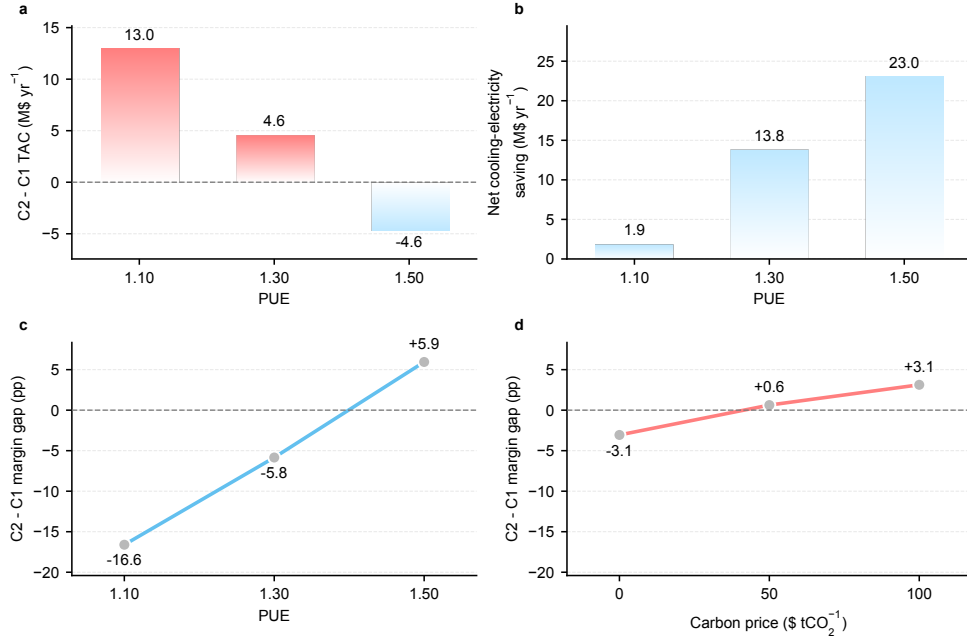


Fig. S2 Cooling-side response of the absorption configuration at the 2023 ATB-Mid baseline. ATB-Mid denotes the mid-range National Laboratory of the Rockies Annual Technology Baseline reactor-capital case; TAC, total annualized cost; PUE, power usage effectiveness; VCC, vapor-compression chiller. (a) Case 2 minus Case 1 TAC difference across full-load PUE; positive is a cogeneration penalty, negative a saving. (b) Net cooling-electricity saving from absorption, measured as gross avoided VCC electricity cost less Case 2 VCC backup/top-up, before fixed absorption capital and turbine-opportunity costs. (c) The same difference expressed as a grid-cost-margin gap across full-load PUE. (d) The margin gap across carbon price. Absorption becomes economic only when the competing electric chiller is inefficient (high PUE) or carbon is priced.

Supplementary Note 9: Scenario run ledger

The baseline and eight sensitivity groups, comprising the four baseline case runs and the 96 sensitivity runs evaluated in this study, are listed in Supplementary Table S7.

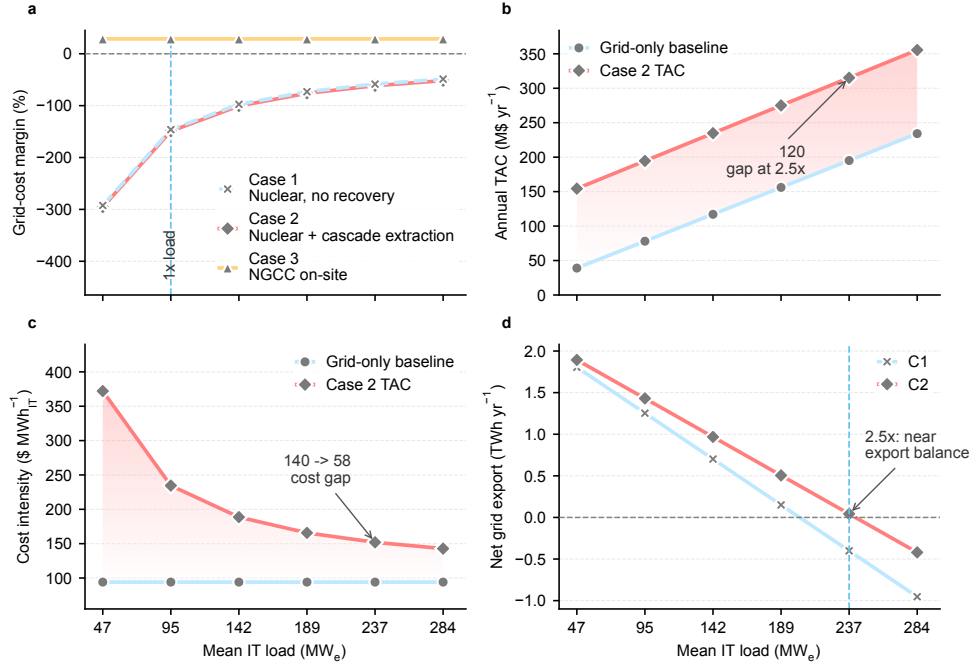


Fig. S3 Data-center size matching sensitivity (G8). IT denotes information technology; NGCC, natural gas combined cycle; TAC, total annualized cost. The IT-load trace and data-center-side cooling and NGCC capacities are scaled from $0.5\times$ to $3.0\times$ while the BWRX-300 reactor remains fixed at 270 MW_e net. (a) TAC-normalized grid-cost margin for the on-site cases against the matching grid-only baseline. (b) Case 2 TAC against the matching grid-only TAC, with the shaded area indicating the annual dollar gap. (c) IT-energy-normalized cost intensity; the shaded area is the excess Case 2 cost over grid supply. (d) Net grid export for the two nuclear cases; positive values indicate merchant export and negative values indicate net grid import. Arrow labels highlight the $2.5\times$ load reference discussed in the text.

Table S2 Input data sources, baseline values and units. ERCOT denotes Electric Reliability Council of Texas; LMP, locational marginal price; IT, information technology; NLR ATB, National Laboratory of the Rockies Annual Technology Baseline; FOAK, first-of-a-kind; NOAK, nth-of-a-kind; O&M, operations and maintenance; VCC, vapor-compression chiller; BESS, battery energy storage system; NGCC, natural gas combined cycle; WACC, weighted-average cost of capital; PTC, production tax credit.

Quantity	Baseline value or range	Resolution	Source
ERCOT day-ahead LMP, Houston Hub	yearly mean \$28–70 MWh _e ⁻¹ (2022–2024)	hourly	gridstatus.io
ERCOT average carbon intensity, ERCOT system	320–351 g CO ₂ kWh _e ⁻¹ annual mean (hourly average emission factor)	hourly	EIA-930 + United Nations Economic Commission for Europe 2022
Delivered natural gas (Henry Hub + basis)	\$2.35–6.53 MMBtu ⁻¹ (annual mean)	hourly	U.S. Energy Information Administration Henry Hub spot
Houston wet-bulb temperature	−0.5 to 28 °C annual range	hourly	Open-Meteo historical
Data-center IT load	142.4 MW _e peak; 94.7 MW _e mean	hourly	Lawrence Berkeley National Laboratory hyperscale benchmark
BWRX-300 capital cost	\$7,615 kW _e ⁻¹ Moderate; \$14,700 kW _e ⁻¹ FOAK; \$2,250 kW _e ⁻¹ NOAK	scalar	NLR ATB 2024 ¹¹
BWRX-300 fixed O&M	\$121 kW _e ⁻¹ yr ⁻¹	scalar	NLR ATB 2024
BWRX-300 capacity factor	0.92 (fleet average)	scalar	U.S. Energy Information Administration Electric Power Monthly Table 6.07.B ²²
BWRX-300 lifecycle emissions	12 g CO ₂ -eq kWh _e ⁻¹	scalar	United Nations Economic Commission for Europe 2022
Absorption chiller capital cost	\$450–1,200 kW _c ⁻¹ (\$750 baseline)	scalar	American Society of Heating, Refrigerating and Air-Conditioning Engineers Technical Committee 8.3 + vendor literature
VCC capital cost	\$250 kW _c ⁻¹	scalar	American Society of Heating, Refrigerating and Air-Conditioning Engineers typical centrifugal
Li-ion BESS capital cost	\$405 kWh ⁻¹ + \$1,620 kW ⁻¹	scalar	NLR ATB 2024
NGCC capital cost	\$1,330 kW _e ⁻¹	scalar	NLR ATB 2024 F-class Moderate
NGCC direct CO ₂	360 g CO ₂ kWh _e ⁻¹ + 60 g upstream CH ₄ (CO ₂ -eq)	scalar	U.S. Energy Information Administration + Alvarez et al. (2018)
WACC	6.7% baseline; {5, 6.7, 10}% in scenario group G7	scalar	Massachusetts Institute of Technology nuclear-finance survey
Asset amortization life	20 yr reactor and turbine; 25 yr chillers; 30 yr NGCC; 15 yr battery	scalar	NLR ATB 2024
Nuclear PTC (Section 45U)	\$15 MWh _e ⁻¹ full credit; phaseout \$25–43.75 MWh _e ⁻¹	scalar	26 U.S.C. 45U; U.S. Internal Revenue Service Form 7213

Table S3 Detailed equipment cost and performance parameters used in the optimization. CAPEX denotes overnight capital expenditure; FOM, fixed operations and maintenance; VOM, variable operations and maintenance; VCC, vapor-compression chiller; BESS, battery energy storage system; NGCC, natural gas combined cycle; HHV, higher heating value; ERCOT, Electric Reliability Council of Texas; PCC, point of common coupling. Each asset’s CAPEX is annualized over its own listed lifetime through an asset-specific capital recovery factor; the battery additionally carries a 5% end-of-life salvage credit.

Asset	CAPEX	FOM (yr^{-1})	VOM	Lifetime	Other technical parameters
BWRX-300 reactor (Cases 1, 2)	$\$7,615 \text{ kW}_e^{-1}$	$\$121 \text{ kW}_e^{-1} \text{ yr}^{-1}$	$\$3.50 \text{ MWh}_e^{-1}$	20 yr	870 MW_{th} , 270 MW_e net; $\eta_r = 0.345$; min-load fraction 0.50; ramp $1\% \text{ min}^{-1}$; capacity factor 0.92; fuel $\$6 \text{ MWh}_{\text{th}}^{-1}$; lifecycle emissions $12 \text{ g CO}_2 \text{ kWh}_e^{-1}$
Cascaded steam turbine (Cases 1, 2)	bundled into reactor overnight capital cost	bundled	bundled	20 yr	$\eta_r = 0.345$ rated at zero extraction; auxiliary load 10%; Willans slope $\alpha_w =$ $0.083 \text{ MW}_e \text{ MW}_{\text{th}}^{-1}$ nameplate cap COP 1.30; design $T_{\text{wb},0} = 26^\circ\text{C}$, COP 1.10; de-rating 0.015°C^{-1} ; crystallization $T_{\text{cry}} = 32^\circ\text{C}$ cooling-water inlet; tower approach 5 K; parasitic $\beta_a =$ $0.020 \text{ kW}_e \text{ kW}_c^{-1}$; 160 MW_c nameplate
Vapor- compression chiller (Cases 0, 1, 2, 3)	$\$250 \text{ kW}_c^{-1}$	$\$8 \text{ kW}_c^{-1} \text{ yr}^{-1}$	$\$0.50 \text{ MWh}_c^{-1}$	25 yr	water-cooled centrifugal; nameplate COP 3.5; Houston- derated COP 3.2; 160 MW_c in all cases
Li-ion BESS (G3)	$\$405 \text{ kWh}^{-1}$ + $\$1,620 \text{ kW}^{-1}$	$\$25 \text{ kW}^{-1} \text{ yr}^{-1}$	$\$0.50 \text{ MWh}^{-1}$	15 yr	100 MWh, 50 MW (C/2); round-trip $\eta_{\text{rt}} = 0.85$; self- discharge $0.1\% \text{ day}^{-1}$; state-of-charge bounds [0.05, 0.95]
NGCC (Case 3)	$\$1,330 \text{ kW}_e^{-1}$	$\$29.3 \text{ kW}_e^{-1} \text{ yr}^{-1}$	$\$2.65 \text{ MWh}_e^{-1}$	30 yr	200 MW_e F-class; HHV $\eta_{\text{hhv}} = 0.495$; direct CO_2 360 g kWh_e^{-1} ; upstream CH_4 60 g kWh_e^{-1} ($\text{CO}_2\text{-eq}$); Henry Hub + Houston Ship Channel basis $\$0.10 \text{ MMBtu}^{-1}$
ERCOT interconnection (Cases 0, 1, 2)	not capitalized	N/A	N/A	N/A	300 MW PCC bidirectional; flat substation interconnection in baseline

Table S4 Market inputs and on-site economics by operating year. ERCOT denotes Electric Reliability Council of Texas; LMP, locational marginal price; TAC, total annualized cost; NGCC, natural gas combined cycle. ERCOT LMP is the annual-mean Houston-zone day-ahead price; delivered gas is the annual-mean Henry Hub plus Houston Ship Channel basis. The grid-cost margin M is positive when the configuration is cheaper than grid procurement.

Quantity	Unit	2022	2023	2024
ERCOT day-ahead LMP	\$ MWh _e ⁻¹	70	57	28
Delivered natural gas	\$ MMBtu ⁻¹	6.53	2.63	2.35
Grid-only (Case 0) TAC	\$M yr ⁻¹	86.2	78.1	36.9
NGCC (Case 3) TAC	\$M yr ⁻¹	87.1	55.7	53.6
NGCC grid-cost margin M	%	-1.1	+28.7	-45.1

Table S5 Marginal annual value of adding the absorption chiller to a nuclear-only plant (Case 2 minus Case 1), at the 2023 ATB-Mid baseline (full-load PUE 1.35) and across the cooling-efficiency sweep. ATB-Mid denotes the mid-range National Laboratory of the Rockies Annual Technology Baseline reactor-capital case; PUE, power usage effectiveness; VCC, vapor-compression chiller; O&M, operations and maintenance. Positive values reduce Case 2 cost relative to Case 1; negative values increase it. All entries are \$M yr⁻¹.

Component	Baseline	PUE 1.10	PUE 1.30	PUE 1.50
Gross avoided VCC electricity	+18.24	+5.25	+15.76	+26.26
VCC backup and top-up	-2.26	-3.31	-1.95	-3.25
Absorption capital + fixed O&M	-13.22	-13.22	-13.22	-13.22
Willans extraction opportunity cost	-4.15	-1.46	-4.15	-4.15
Production tax credit forgone	-0.62	-0.03	-0.62	-0.62
Direct site-water cost	-0.04	-0.01	-0.04	-0.04
Residual reconciliation	-0.35	-0.18	-0.35	-0.35
Net value of absorption	-2.39	-12.96	-4.56	+4.64

Table S6 Data-center size-matching results for Case 2 (group G8) at the 2023 ATB-Mid baseline. ATB-Mid denotes the mid-range National Laboratory of the Rockies Annual Technology Baseline reactor-capital case; IT, information technology; TAC, total annualized cost. The reactor is held at 270 MW_e net while the IT load and data-center-side capacities scale with the multiplier. M is the grid-cost margin against the matched grid-only baseline; net export is annual grid sales minus purchases.

Load multiplier	0.5	1.0	1.5	2.0	2.5	3.0
Case 2 TAC (\$M yr ⁻¹)	154.3	194.6	234.8	275.0	315.3	355.5
Grid-only TAC (\$M yr ⁻¹)	39.0	78.1	117.1	156.1	195.1	234.2
Annual gap (\$M yr ⁻¹)	115.3	116.5	117.7	118.9	120.1	121.3
Grid-cost margin M (%)	-295	-149	-101	-76	-62	-52
Net grid export (TWh yr ⁻¹)	+1.89	+1.43	+0.97	+0.51	+0.04	-0.42
Cost intensity (\$ MWh _{IT} ⁻¹)	372	235	189	166	152	143
Excess over grid (\$ MWh _{IT} ⁻¹)	278	140	95	72	58	49

Table S7 Baseline and sensitivity scenario run ledger. C0–C3 denote Cases 0–3; PUE, power usage effectiveness; ATB-Mid, mid-range National Laboratory of the Rockies Annual Technology Baseline reactor-capital case; CAPEX, overnight capital expenditure; WACC, weighted-average cost of capital; ERCOT, Electric Reliability Council of Texas; BESS, battery energy storage system; SMR, small modular reactor; FOAK, first-of-a-kind; NOAK, nth-of-a-kind; IT, information technology. Cells marked -- indicate that the case is not included in that scenario group.

Group	Sweep	Levels	Runs	C0	C1	C2	C3
G0	Baseline	2023, full-load PUE 1.35, ATB-Mid CAPEX, WACC 6.7%	4	✓	✓	✓	✓
G1	Effective PUE	1.10, 1.30, 1.50	6	–	✓	✓	–
G2	ERCOT year	2022, 2023, 2024	12	✓	✓	✓	✓
G3	BESS	on, off	8	control	✓	✓	control
G4	SMR CAPEX	FOAK, ATB-Mid, NOAK	6	–	✓	✓	–
G5	SMR × Abs. CAPEX	5 × 5 grid	25	–	–	✓	–
G6	Carbon price	\$0, \$50 and \$100 tCO ₂ ^{−1}	12	✓	✓	✓	✓
G7	WACC	5, 6.7, 10%	3	–	–	✓	–
G8	Data-center size	0.5–3.0× IT load	24	✓	✓	✓	✓
Total			100				

Note: In G3, control denotes duplicated control rows retained in the run ledger; BESS dispatch is optimized only for Cases 1 and 2 because Cases 0 and 3 are closed-form annual cost evaluations.